Jinyuan Mao

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Education

Zhejiang University Hangzhou, China
B.S. in Statistics & B.Mgt. in Info. Management and Info. Systems (Dual Degree) *Sep. 2023 – Jun. 2027*

- **GPA:** 4.19/4.3, **Rank:** 1/20
- **Research Interests:** Interactive Visual Generation, Video World Models, Efficient AI

Publications

* denotes equal contribution

[1] **HimmBO: High-Dimensional Interactive Model-Merging Bayesian Optimization for Image Generation**
Jinyuan Mao, Zhihao Shen, Chenxi Liu
Under review at Eurographics 2027 | arXiv

[2] **WorldAttention: An Efficient Attention Architecture for Interactive Video World Models**
Zeyu Zhang*, **Jinyuan Mao***, Dakai An*, Wangbo Zhao, Hanfeng Lu, Jiasheng Tang, Yinghao Yu, Wei Wang, Bohan Zhuang
NeurIPS 2026 submission (review scores: 5/5/5 – Accept/Accept/Accept) | arXiv

[3] **BIFE: Better Interaction, Fewer Errors for Minute-Long Video Generation**
Zeyu Zhang*, **Jinyuan Mao***, Shuning Chang, Yuanyu He, Yizeng Han, Jiasheng Tang, Fan Wang, Bohan Zhuang
SIGGRAPH Asia 2026 Technical Communications | arXiv

[4] **IntuiTF: MLLM-Guided Transfer Function Optimization for Direct Volume Rendering**
Yiyao Wang, Bo Pan, Ke Wang, Han Liu, **Jinyuan Mao**, Yuxin Liu, Minfeng Zhu, Xiuqi Huang, Weifeng Chen, Bo Zhang, Wei Chen
arXiv preprint, 2025 | arXiv

Research Experience

Research Intern, University of Virginia Jul. 2026 – Present
Pixel-space diffusion and its frequency structure; advised by Prof. Sheng Li

- Proposed **High-Resolution Guidance**, which updates low-frequency predictions with a cheap low-resolution model and recomputes guidance only at anchor steps; **2.2**× less sampling compute on 512 px MiniT2I with no loss in PickScore or GenEval, outperforming CFG-Cache at matched cost.
- Designed a pixel-space MoE reaching FID **4.22** on ImageNet vs. 4.59 for dense JiT, with 60% of JiTMoE's parameters.

Research Intern, PixAI May 2026 – Sep. 2026
Interactive image generation; advised by Prof. Chenxi Liu (University of Victoria) *Hangzhou, China*

- As first author and sole student researcher, proposed **HimmBO** [1], a two-stage preferential Bayesian optimization framework that scales interactive adapter (LoRA) merging from a few dozen to **1080 adapters**, with a level-set screening stage that needs no image generation during interaction.
- Designed a style-aware kernel that transfers preference information across adapters; reached a given similarity with **under 1/10** of the inference required by prior work, and stayed stable as the library grew while baselines degraded.

Research Intern, ZIP Lab, Zhejiang University Oct. 2025 – Jun. 2026
Efficient video generation and world models; advised by Prof. Bohan Zhuang *Hangzhou, China*

- **BIFE** [3]: co-designed a semantic sparse KV cache that retrieves relevant history instead of keeping a sliding window, preserving long-range interactions in minute-long video generation.
- **WorldAttention** [2]: found that attention heads in video diffusion transformers have highly heterogeneous sparsity, and designed **Hybrid Sparse Attention**, pairing a linear global branch with head-adaptive block-sparse attention; implemented it end to end, achieving a **2.21**× inference speedup at matched quality.

Supercomputing Experience

Member, ZJU Supercomputing Team Oct. 2024 – May 2026
Student cluster competitions and HPC application optimization *Zhejiang University*

- **SC25 IndySCC** (1st place globally): built, parallelized, and tuned the SST architecture simulator, and searched for the best-performing simulation configurations.
- **ICT Competition** (1st Place, National Finals): ported a multimodal LLM (InternVL2.5) to an NPU-based distributed training stack, achieving a **1.15**× training speedup.
- **PAC 2025** (National 2nd Prize): sped up floating-point matrix kernels **4**× with multi-threading, SME vectorization, and block prefetching.

Other Experience

AI Engineering Intern, Nodesk AI Jun. 2025 – Aug. 2025
Algorithm development *Hangzhou, China*

- Built a smart-reply cross-validation module, improving accuracy by **15%** via fine-tuning; co-authored a patent on a guardrail mechanism combining reranking models with scenario recognition.

Operations & Infrastructure Lead, ZJU X-Lab Jun. 2025 – Present
Student-run innovation organization *Zhejiang University*

- Manage the GPU clusters with Docker-isolated environments for 10+ projects; built a private Overleaf, NAS storage, VPN, and the organization's website.

Honors & Awards

National Scholarship (Top 0.2%), three times 2024, 2025, 2026
1st Place Globally, SC25 IndySCC Student Cluster Competition Nov. 2025
National 2nd Prize, 12th National Parallel Application Challenge (PAC) Aug. 2025
2nd Prize, ASC25 Student Supercomputer Challenge May 2025
1st Place, National Finals, ICT Competition Mar. 2025

Skills

Programming: Python, C/C++, CUDA, TypeScript
ML & HPC: PyTorch, Diffusers, MPI, OpenMP, Slurm, Docker; profiling with Nsight Systems, VTune, perf